Qualcomm Technologies, Inc.

AudioReach Voice Architecture

80-VN500-2 Rev. AC

September 7, 2022

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Revision history

Revision	Date	Description
AA	November 2020	Initial release
AB	March 2021	Added Chapter 6. Applied terminology updates and improved the quality of images.
AC	September 2022	Editorial updates

Note: There is no Rev. I, O, Q, S, X, or Z per Mil. standards.

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1 Introduction

1.1 Purpose

This document explains the voice call architecture and call flows between the HLOS, ADSP, and modem in AudioReach™ based architecture.

1.2 Conventions

Function declarations, function names, type declarations, attributes, and code samples appear in a different font, for example, `cp armcc armcpp`.

Code variables appear in angle brackets, for example, `<number>`.

Commands to be entered appear in a different font, for example, **`copy a:*.* b:`**.

Button and key names appear in bold font, for example, click **Save** or press **Enter**.

Shading indicates content that has been added or changed in this revision of the document.

1.3 Technical assistance

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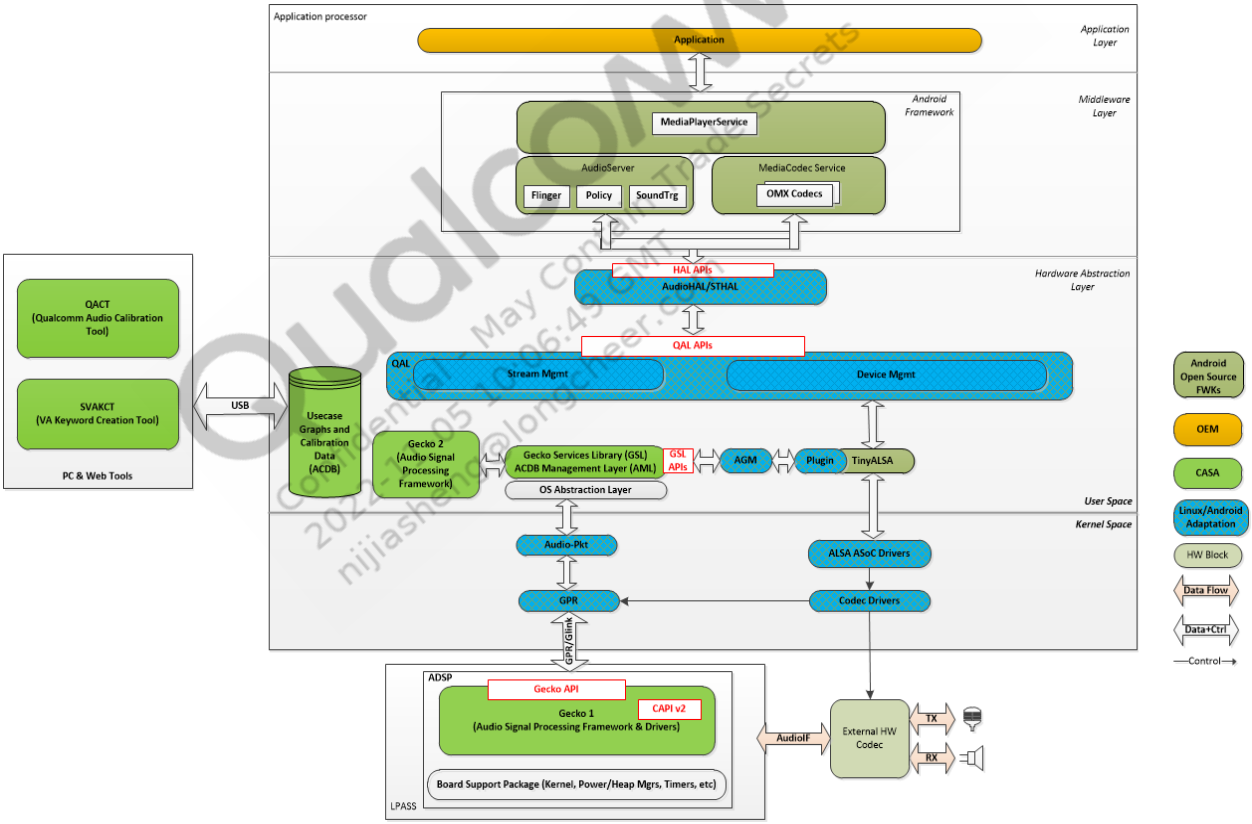
2 High level design

2.1 HLOS architecture

The following diagram shows the AudioReach HLOS architecture.



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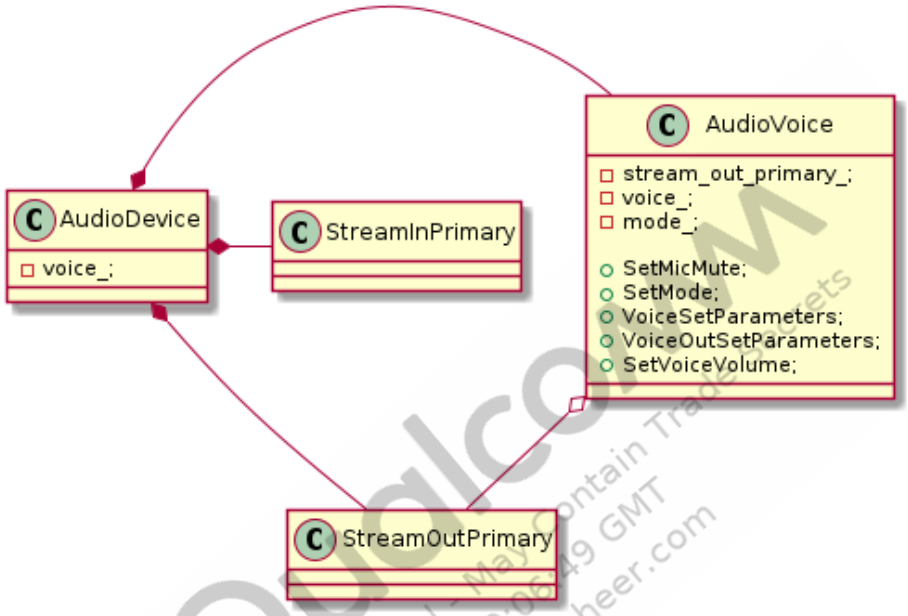
2.2 HAL for voice

2.2.1 High level class diagram

The following is a high level class diagram of the HAL.



2.2.2 AudioDevice class

- The AudioVoice object pointer is member of this class
- Responsibilities include creation and deletion of AudioVoice object
- The AudioVoice object is created only when device is opened for the first time. It is deleted when the last instance of device is closed.
- Handles and forwards voice related commands including:
  - SetMode
  - SetParam
  - SetMicMute
  - SetVolume

2.2.3 AudioVoice class

- Class responsible for handling PAL voice stream operations
- Performs operations based on input from AudioDevice and AudioStream

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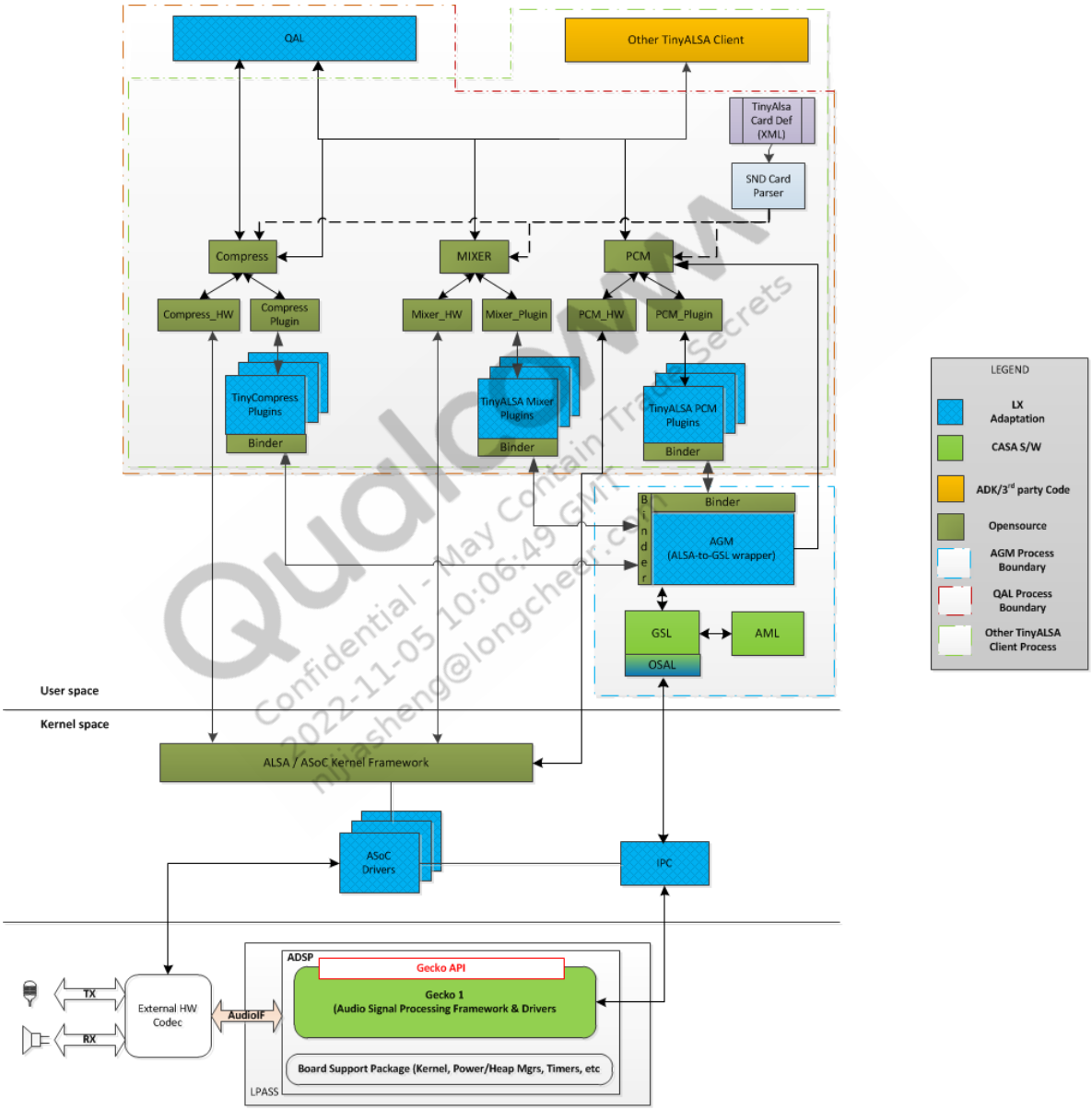
2.3 PAL for voice

2.3.1 PAL interaction diagram

The following diagram shows PAL interactions.



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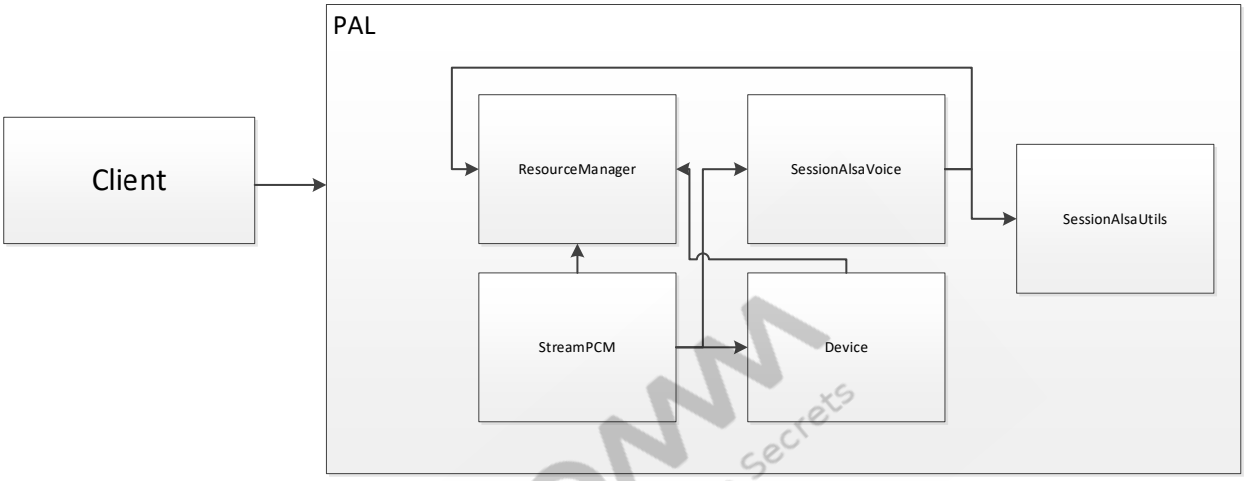
2.3.2 PAL voice blocks

The following diagram shows the PAL voice blocks.



Clients, such as the HAL, interact with PAL via the standard set of APIs. The parameters passed invoke the blocks shown in the diagram that are used for voice call setup and tear down.

2.3.3 PAL voice usage

- Voice call is a unique use case in PAL as it contains both an Rx and Tx stream in one session
- Voice-specific use case implementation can be found in SessionAlsaVoice.cpp
- Voice call is support by PAL stream type:
  - PAL_STREAM_VOICE_CALL
- Voice call can be supported on numerous PAL device pairs:
  - PAL_DEVICE_OUT_HANDSET / PAL_DEVICE_IN_HANDSET_MIC
  - PAL_DEVICE_OUT_SPEAKER / PAL_DEVICE_IN_SPEAKER_MIC
  - PAL_DEVICE_OUT_WIRED_HEADSET / PAL_DEVICE_IN_WIRED_HEADSET
  - PAL_DEVICE_OUT_USB_HEADSET / PAL_DEVICE_IN_USB_HEADSET
  - PAL_DEVICE_OUT_BLUETOOTH_SCO / PAL_DEVICE_IN_BLUETOOTH_SCO_HEADSET

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3 Low level design

3.1 Graph design and GKVs

Each use case is represented by a graph key vector (GKV). In the case of a voice call, each encoder has different modules that need to be configured from the client. To avoid making multiple graphs for each encoder, Placeholder Encoder is used for certain encoders.

3.2 Basic voice call

The basic voice call use case consists of two separate graphs – One for the Rx path and one for the Tx path. These graphs are pushed down to the signal processing framework (SPF) based on the configuration given to PAL. This can come from the HAL or another PAL client.

3.2.1 Rx voice graph

The Rx voice call graph consists of the following subgraphs:

- Stream
- Stream PP
- Device PP
- Device

These subgraphs make up the full Rx voice call processing chain and can be configured and tuned using the QACT™ platform.

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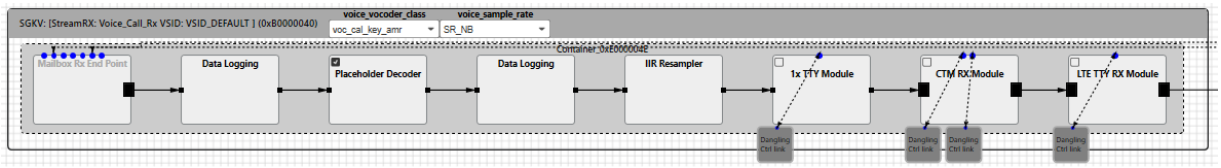
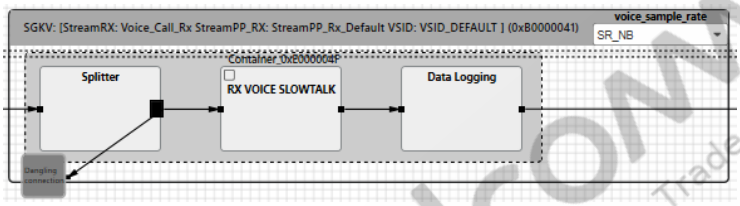
Rx stream leg

The Rx stream leg is shared by all Rx voice graphs as it is generic to all voice sessions.



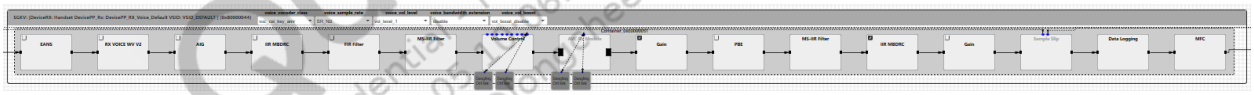
RX stream PP

The Rx stream PP leg is shared by all Rx voice graphs as it is generic to all voice sessions.



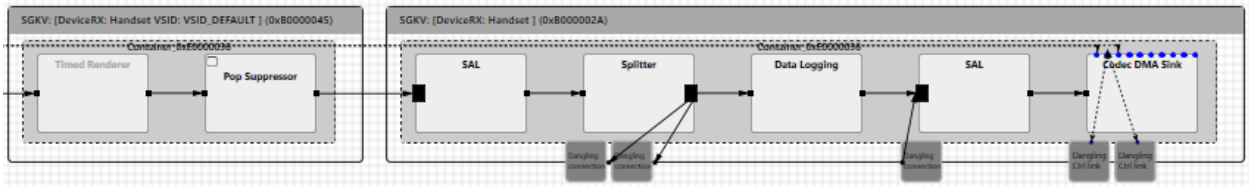
RX device PP

Rx device PP is unique to the device chosen (the following example is handset)



RX device

Rx device is unique to the device chosen. (the following example is handset.



3.2.2 Tx voice graph

The Tx voice call graph consists of the following subgraphs:

- Device
- Device PP
- Stream

These three subgraphs make up the full Tx voice call processing chain and can be configured and tuned using QACT.

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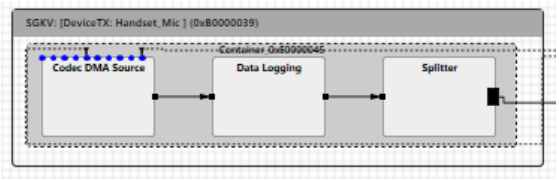
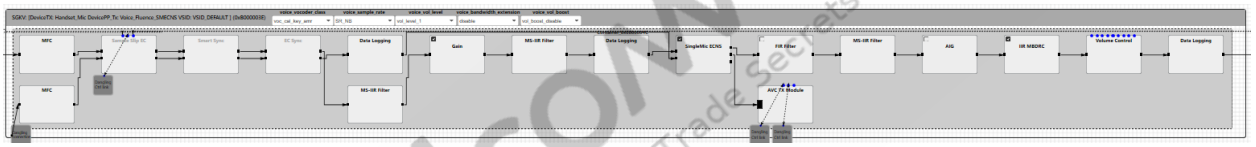
Tx device leg

The TX device leg is unique for all Tx voice call devices (the following example is Handset_mic).



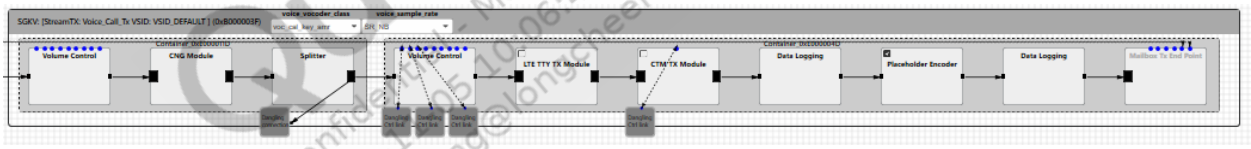
Tx device PP

The Tx device PP chain is unique for all Tx devices as well as all Fluence modes (the following example is Handset_mic, single mic ECNS).



Tx stream

The Tx stream leg is shared across all voice graphs as it is common.



3.3 In-call recording

In-call recording can record a voice stream from a voice call and save the recorded audio into a file on the device. This feature is similar to a regular audio recording except the record tap points are within the voice call stream instead of a microphone endpoint.

Recording can be done in four different modes:

- Uplink recording
- Downlink recording
- Uplink and downlink recording in mono format
- Uplink and downlink recording in stereo format

In-call recording has its own PCM device defined in the card definition xml file. Since the recording source is a voice call, the stream creation does not provide any device endpoint.

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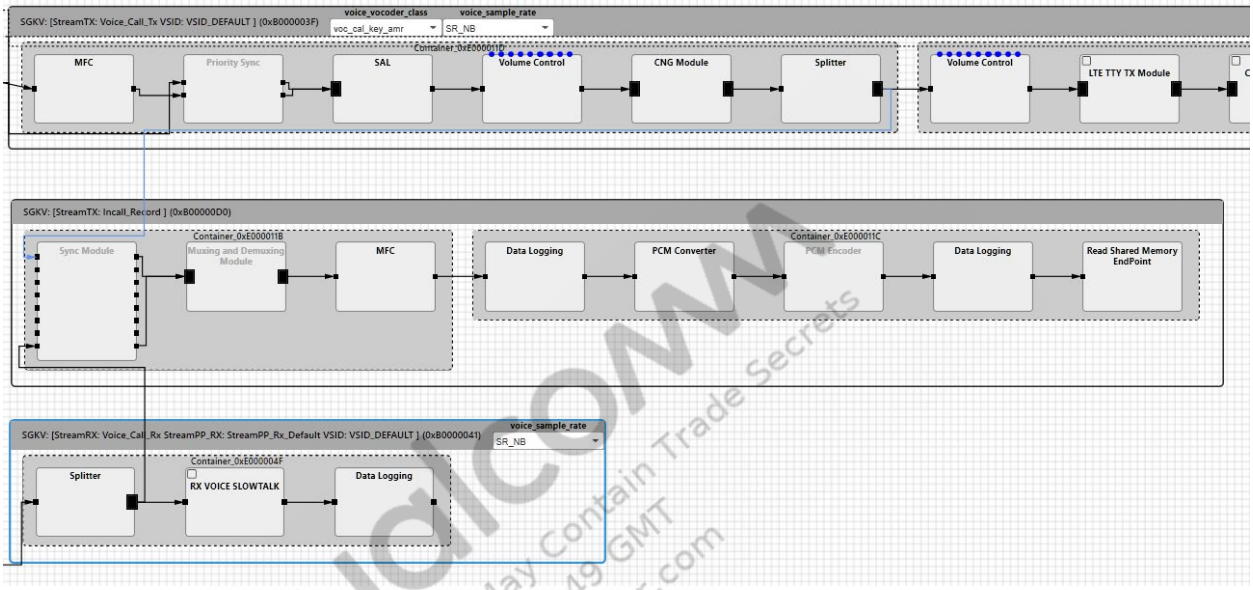
In-call recording subgraph

The in-call recording graph is a separate subgraph from the voice call subgraph and is connected statically. Whenever both use cases are run concurrently, data flows from one subgraph to another. Based on the configuration within the in-call subgraph, the direction of the recording can be controlled.



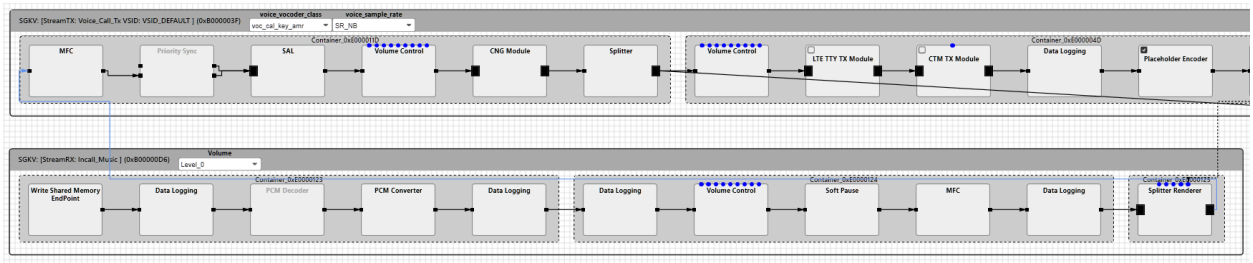
3.4 In-call music delivery

In-call music delivery allows users to start local playback, which is then mixed with the voice call and rendered to the far-end user over the voice call. The music is not rendered on the local (near-end) device.

In-call music delivery has its own PCM device defined in the card definition xml file. Since the music is not played on any local audio device, the stream creation does not provide any device endpoint.

In-call music delivery subgraph

The in-call music delivery graph is a separate subgraph from the voice call subgraph and is connected statically. Whenever both use cases are run concurrently, data flows from one subgraph to another.



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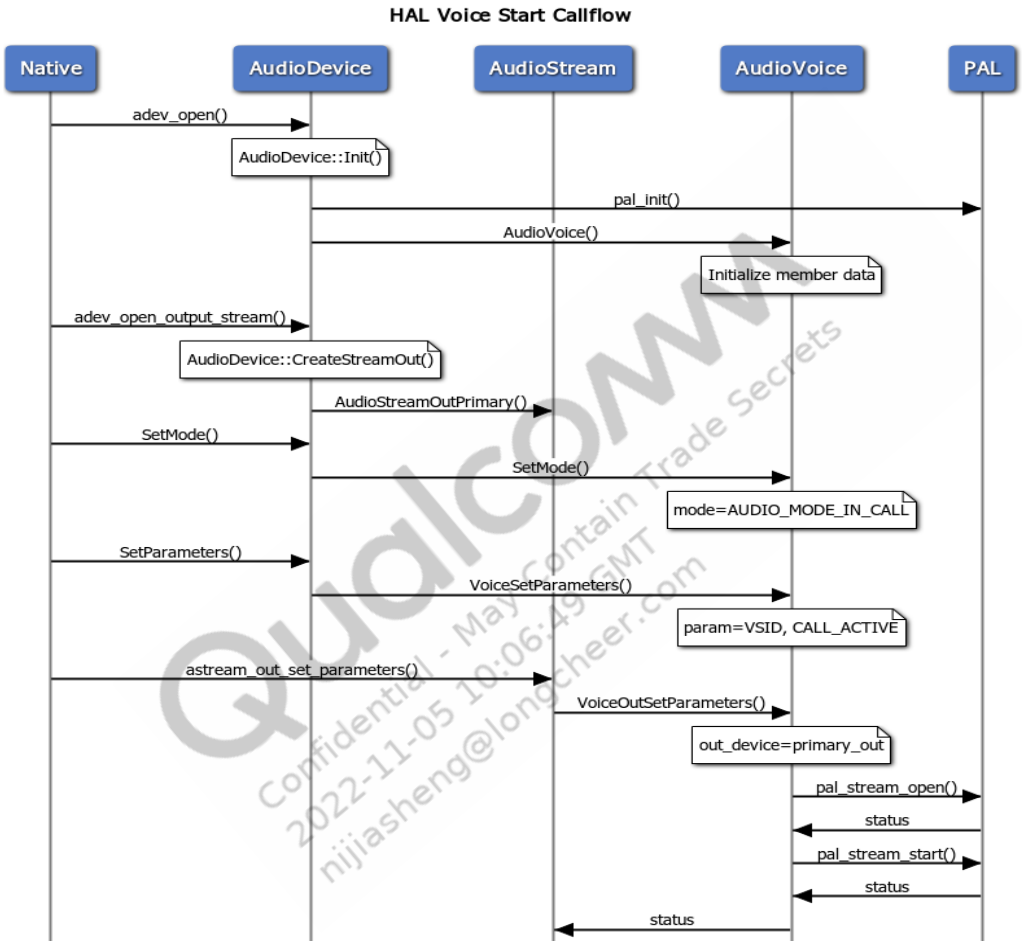
3.5 Call sequence diagrams

3.5.1 Basic voice call



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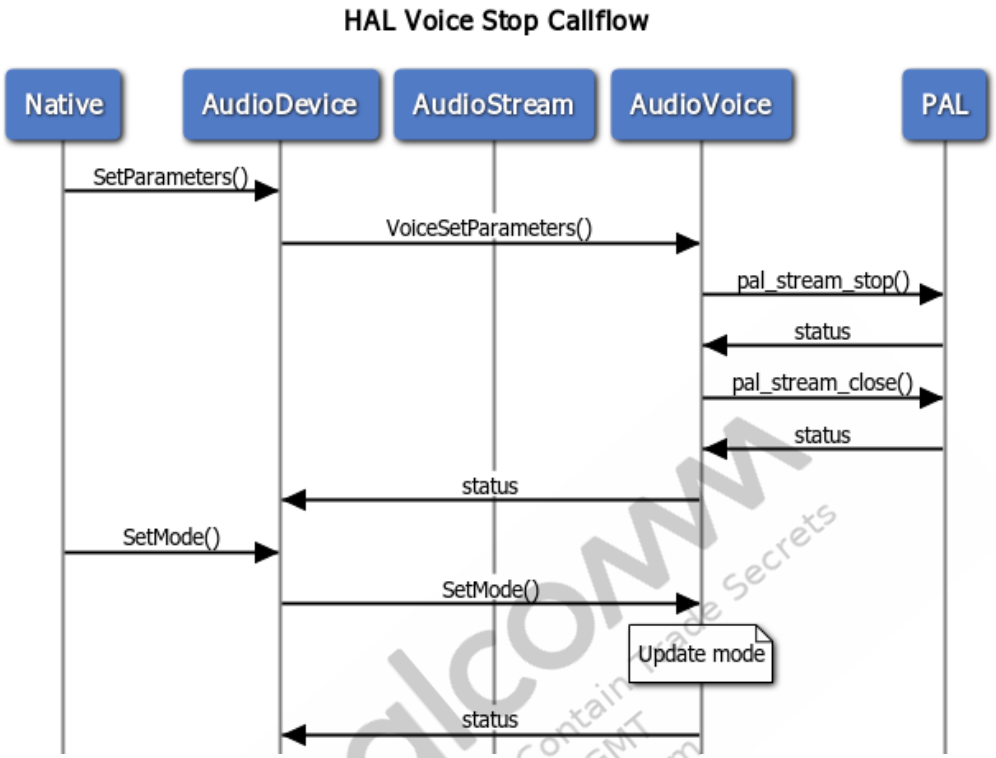

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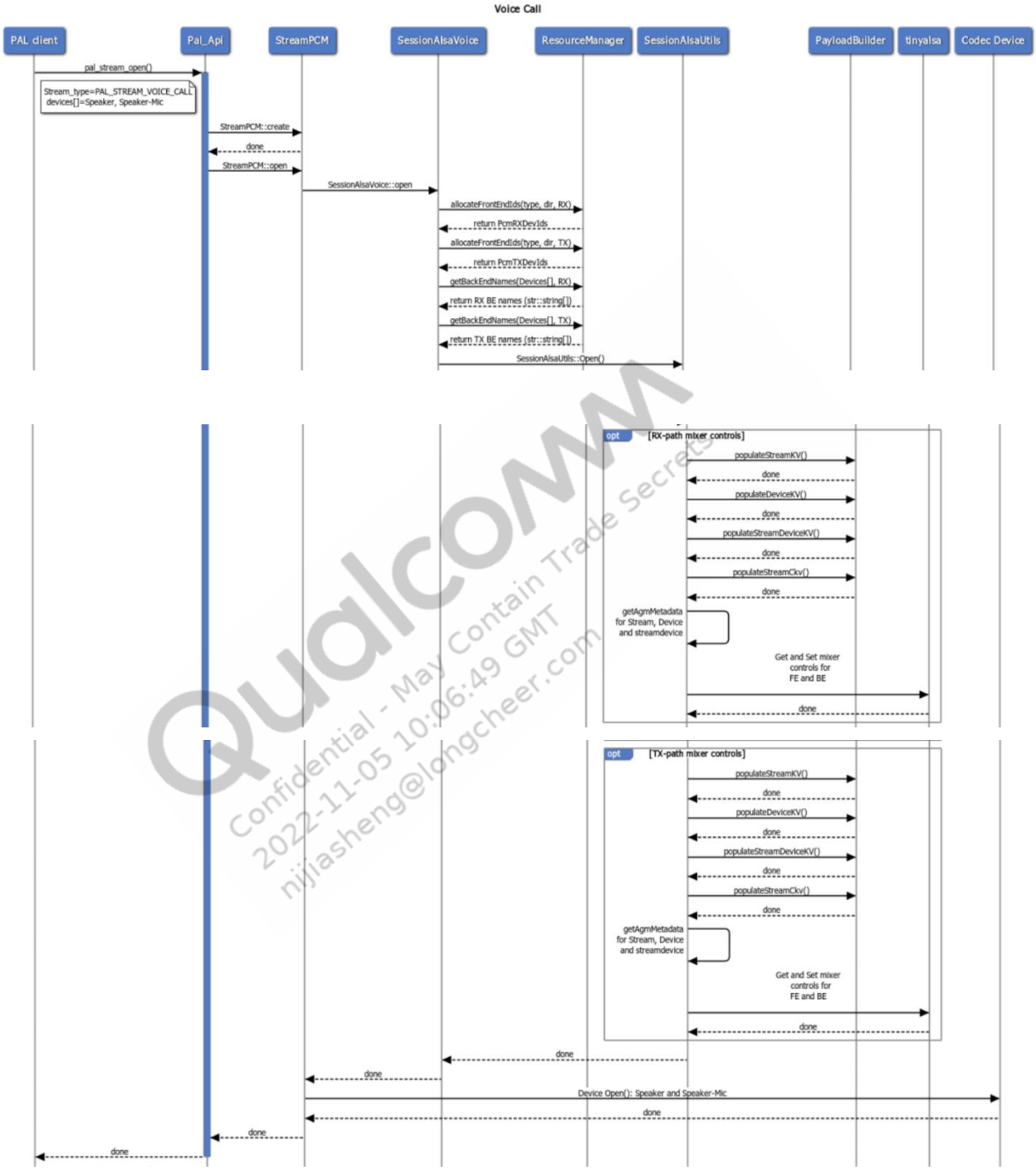

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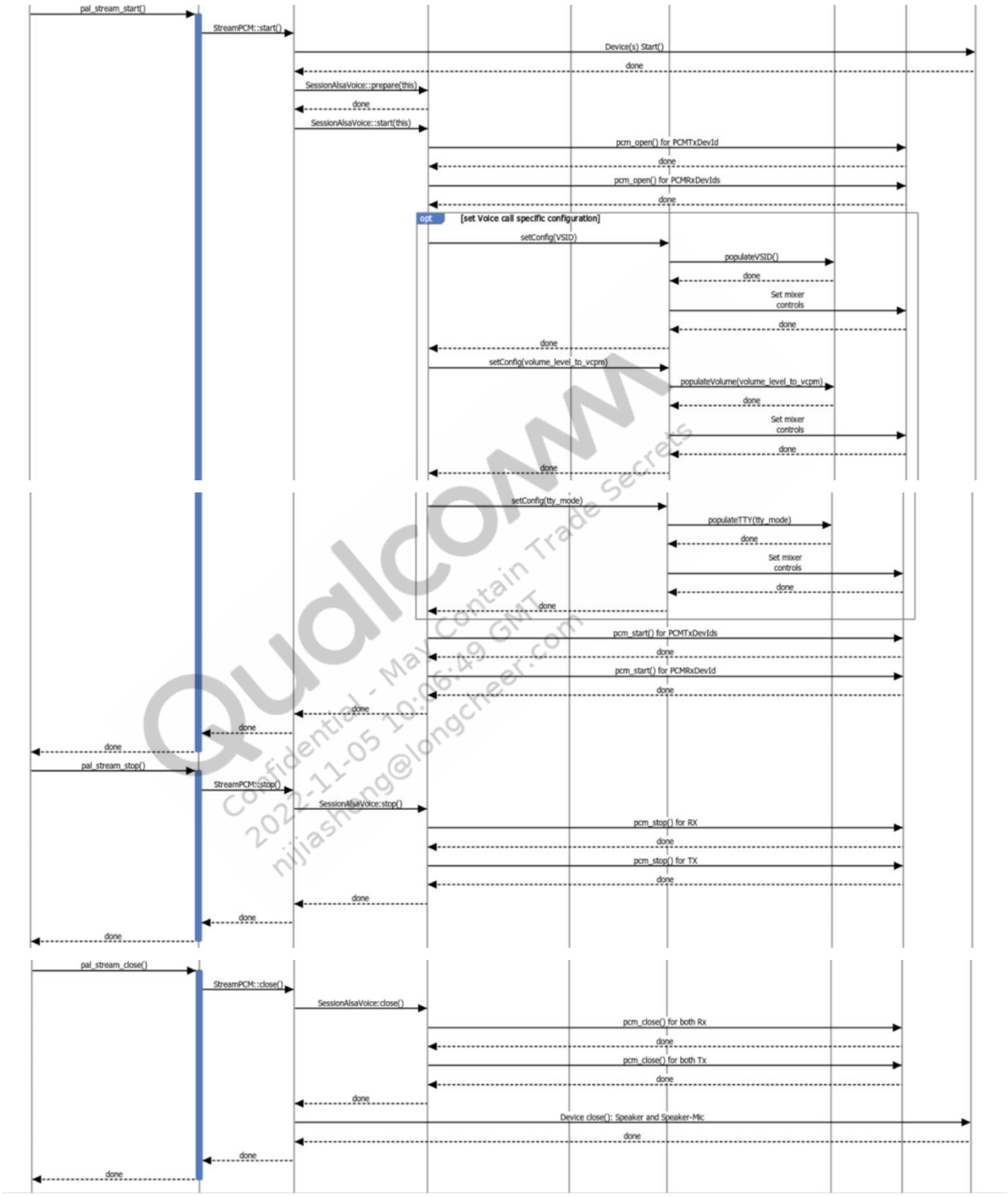

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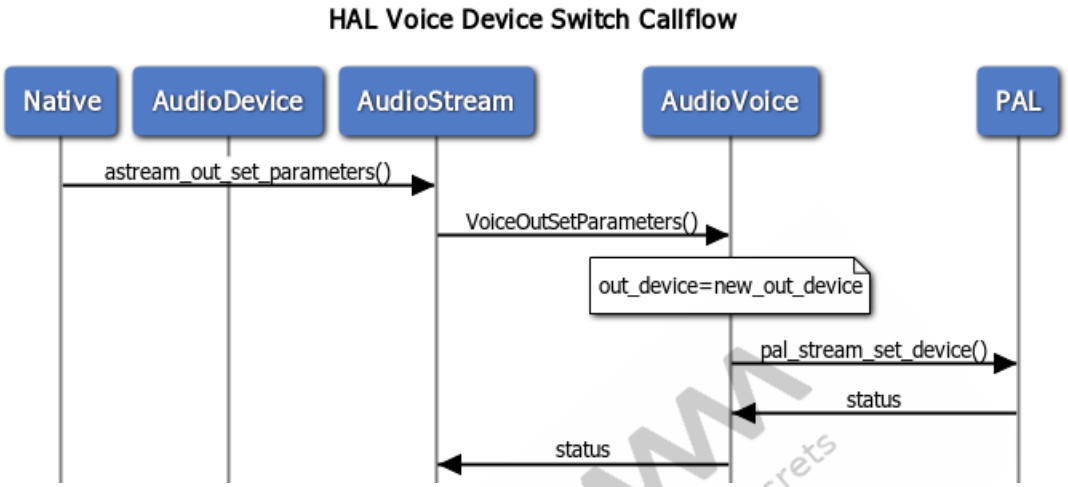
3.5.2 Voice call device switch



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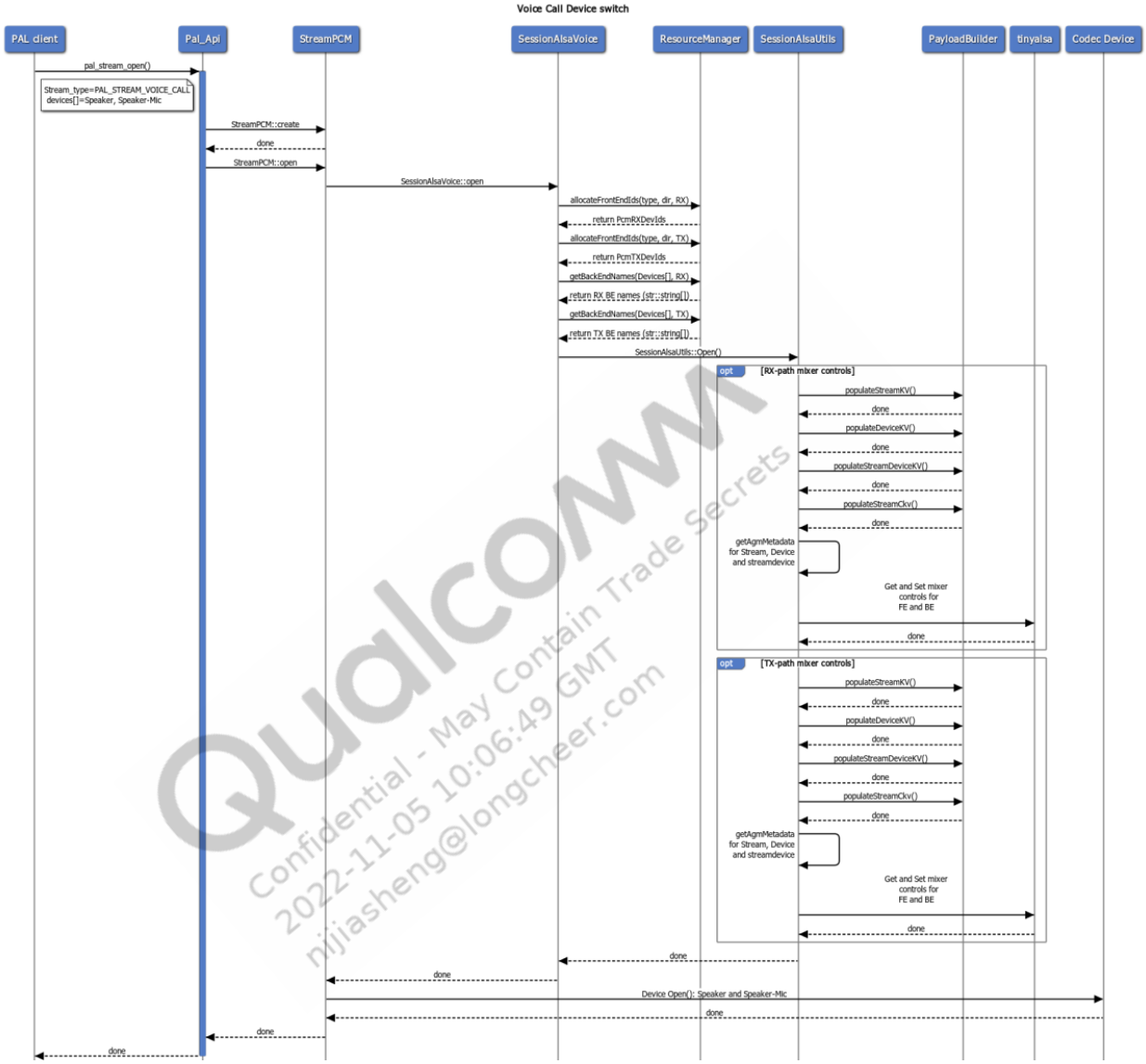

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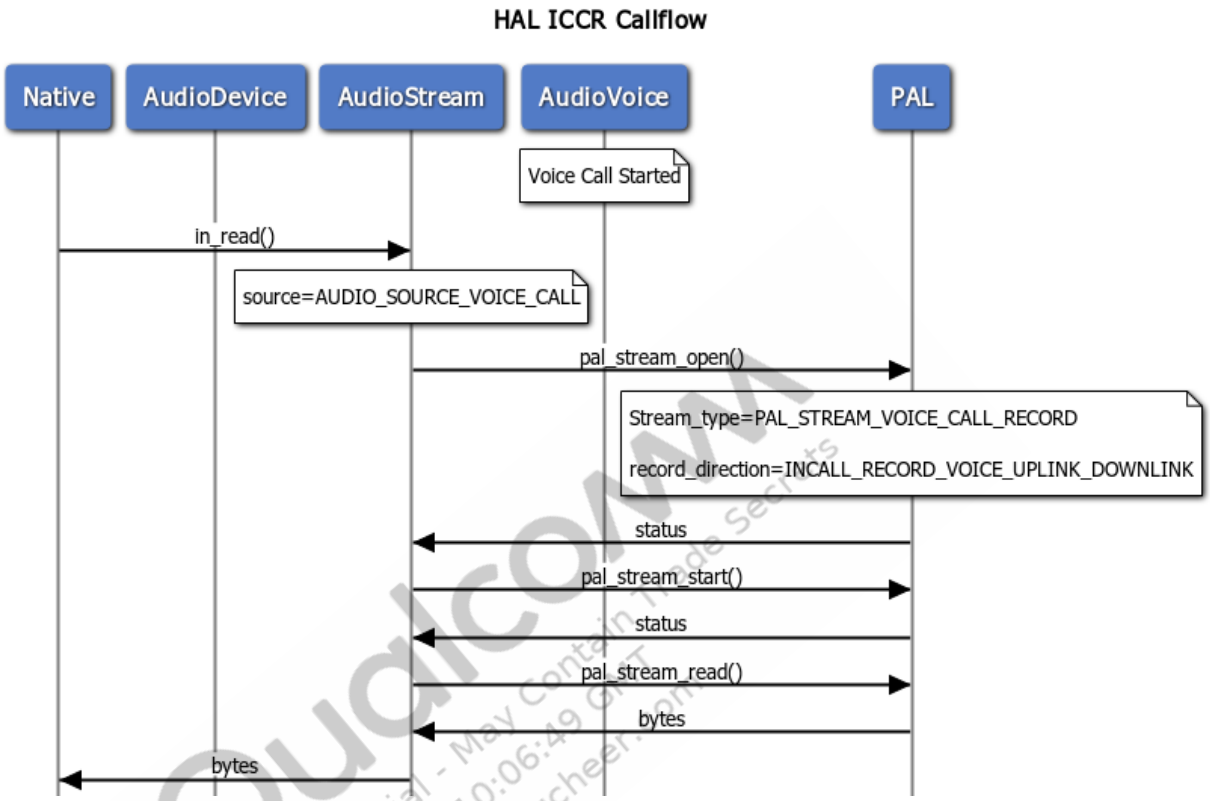
3.5.3 Voice call with in-call record



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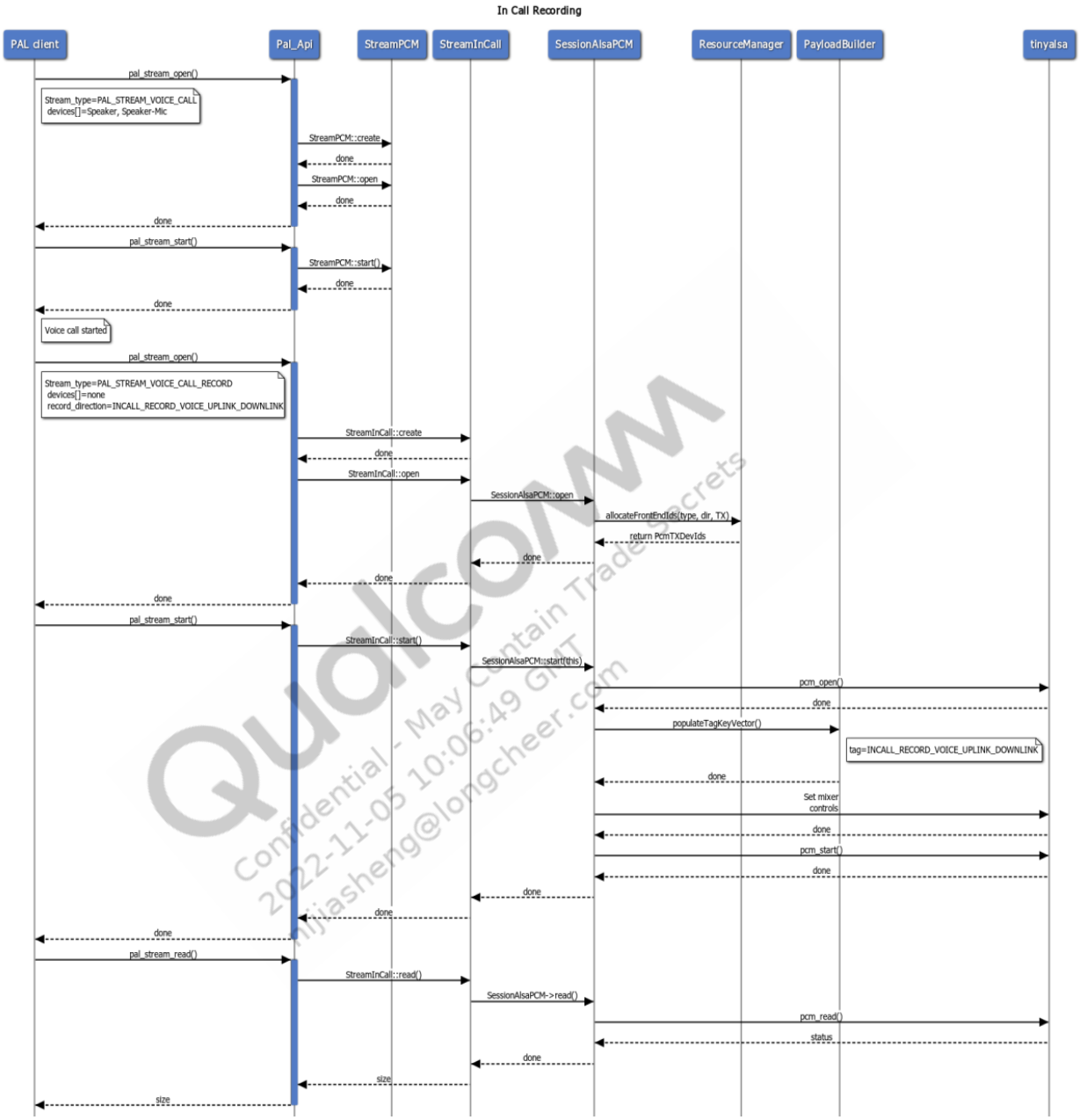

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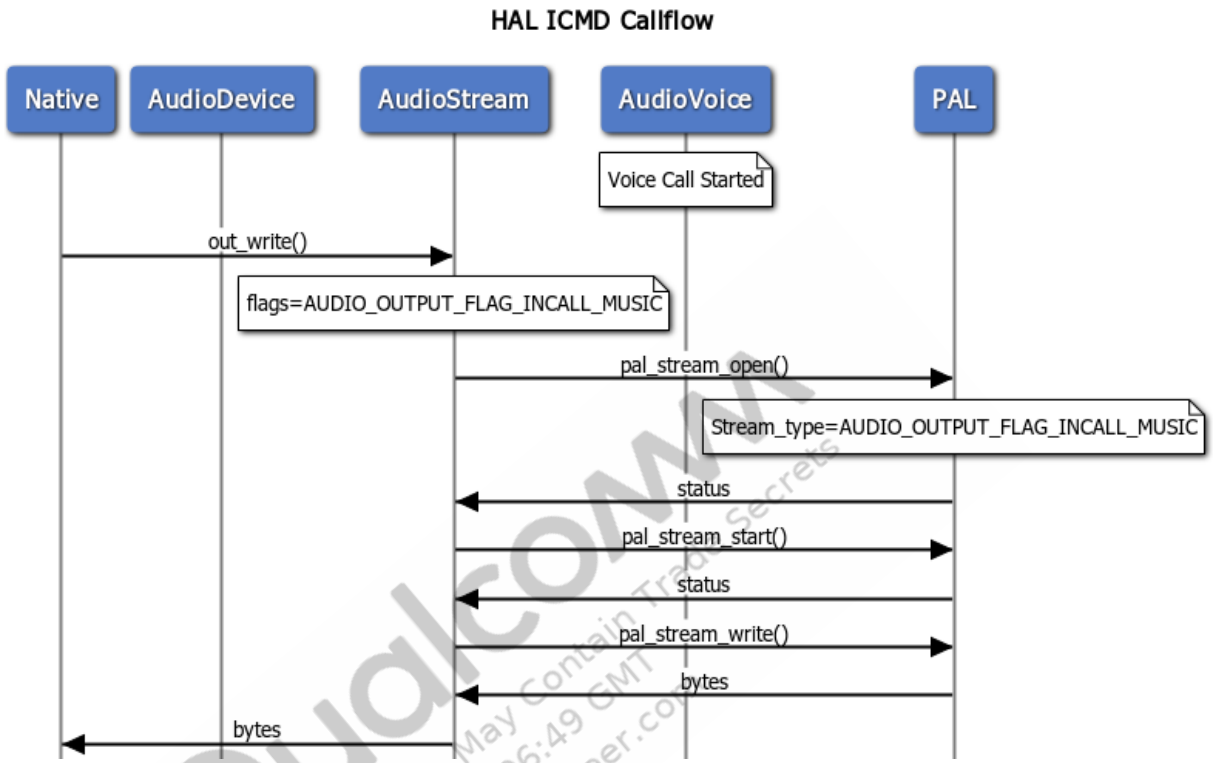
3.5.4 Voice call with in-call music playback



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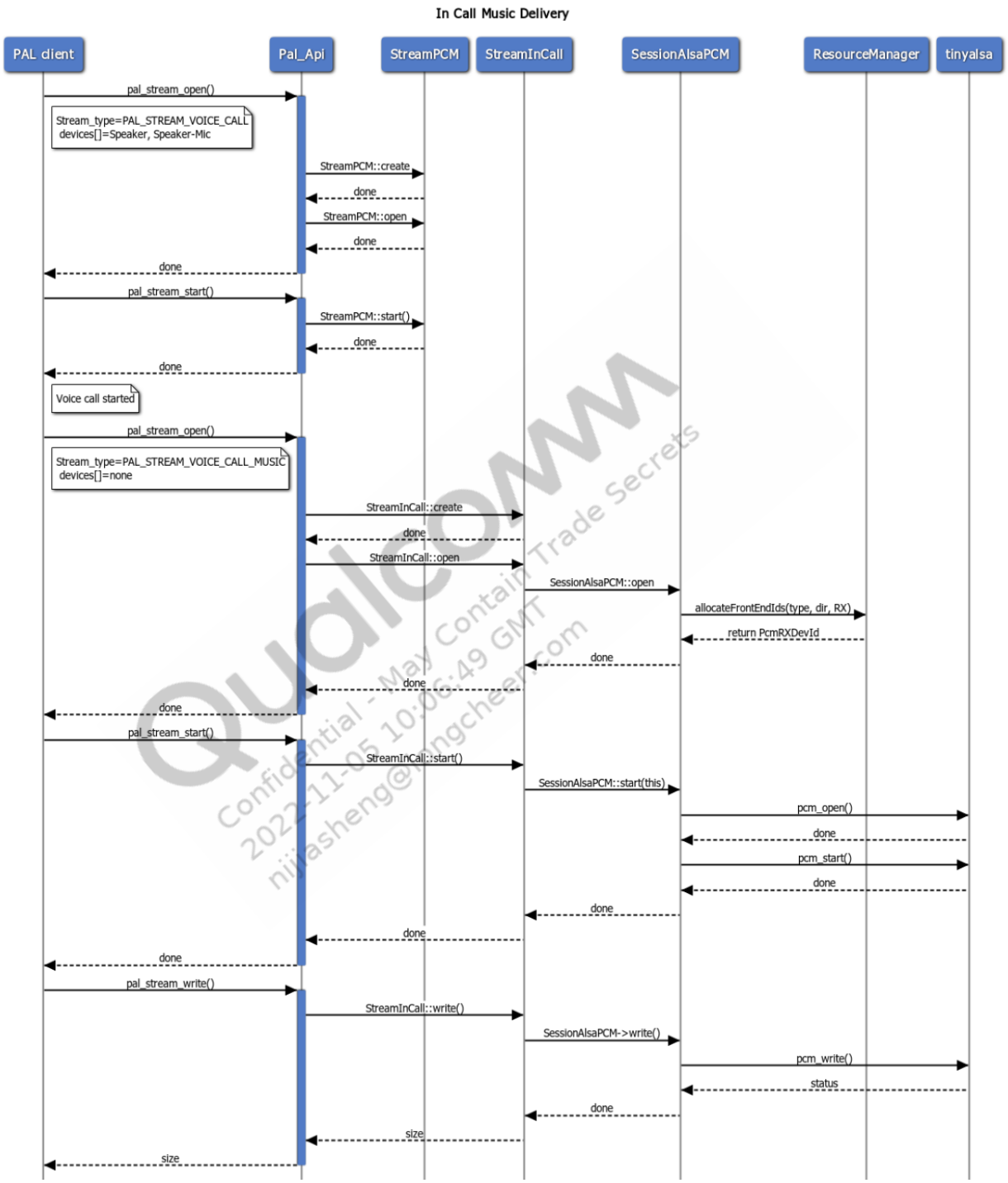

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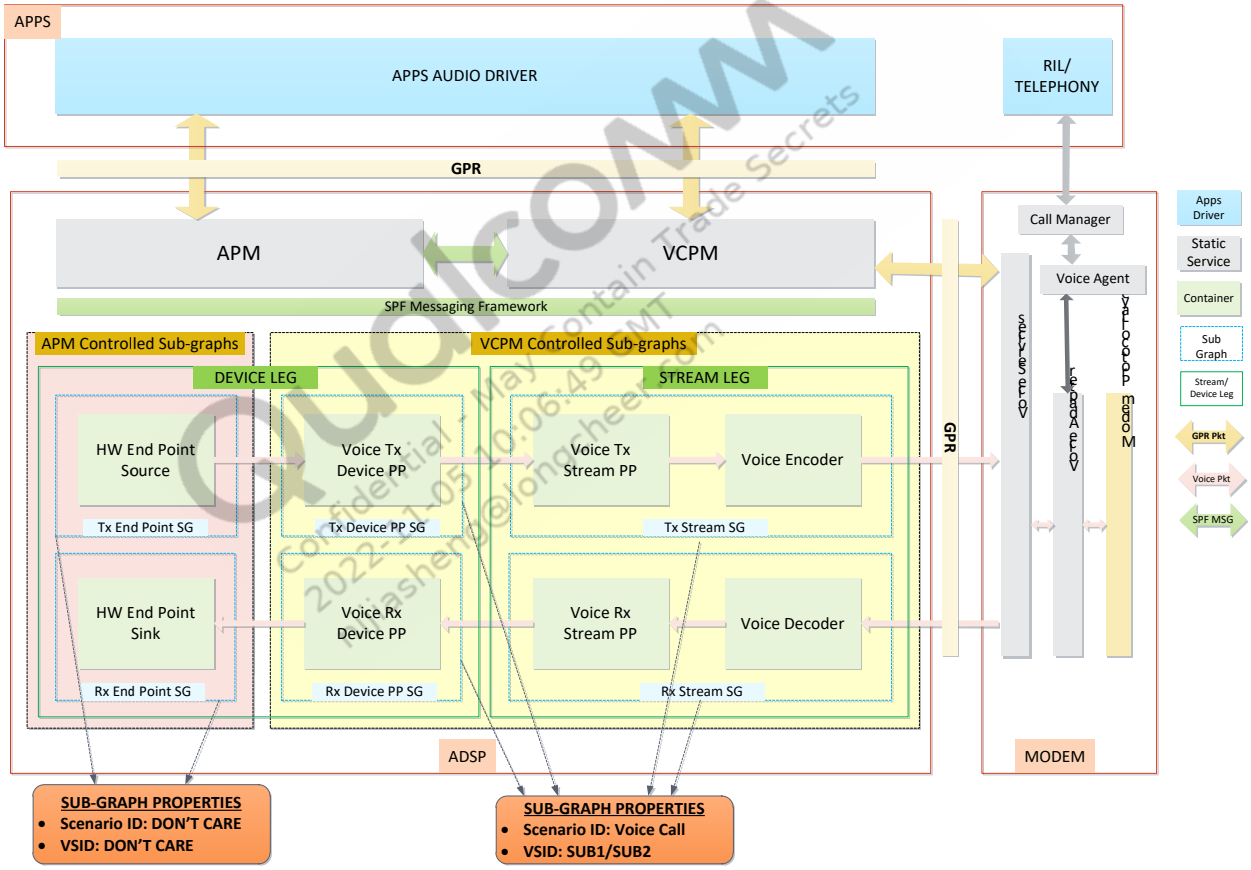
4 SPF voice call framework

4.1 SPF voice architecture

The following diagram shows the SPF voice architecture.



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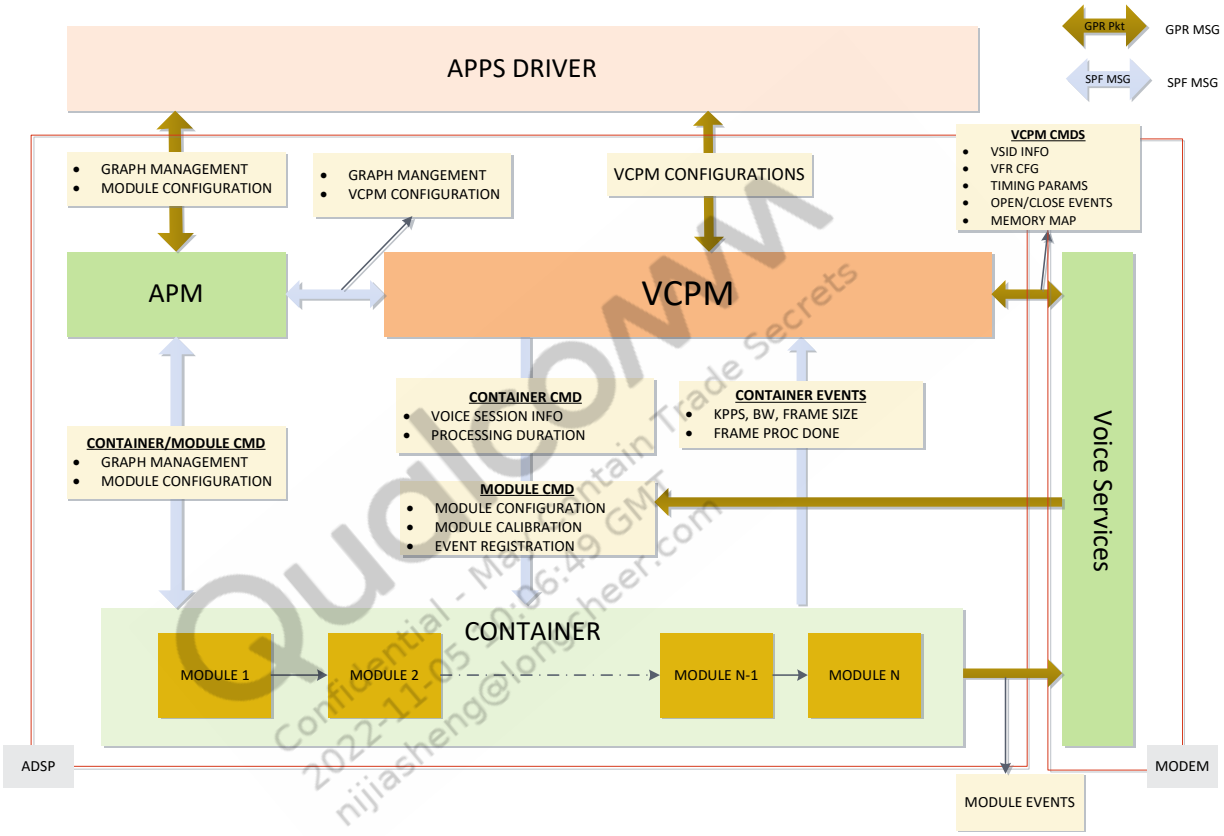
4.2 Voice call proxy manager (VCPM)

VCPM is a static control module that manages the voice call subgraphs. It provides a dual control mechanism for control commands from the apps driver and voice services on the modem and direct interaction with voice services with voice processing modules.

The following diagram shows the VCPM architecture.



4.2.1 Key responsibilities

VCPM responsibilities include:

- State management for voice call subgraphs interfacing with APPS and modem clients for call setup, device switch, and handovers
- Voice call graph setup, UL/DL path delay estimation, Q6 clock and bandwidth voting calculation, and RTD budgeting
- Applying CKV-based module calibration on voice call subgraph modules (for example, based upon vocoder class, sampling rate, etc.)
- Managing the sleep/wakeup cycle of voice call subgraphs for the VoLTE CDRX40 use case
- Managing island entry/exit for the VoLTE LPI use case

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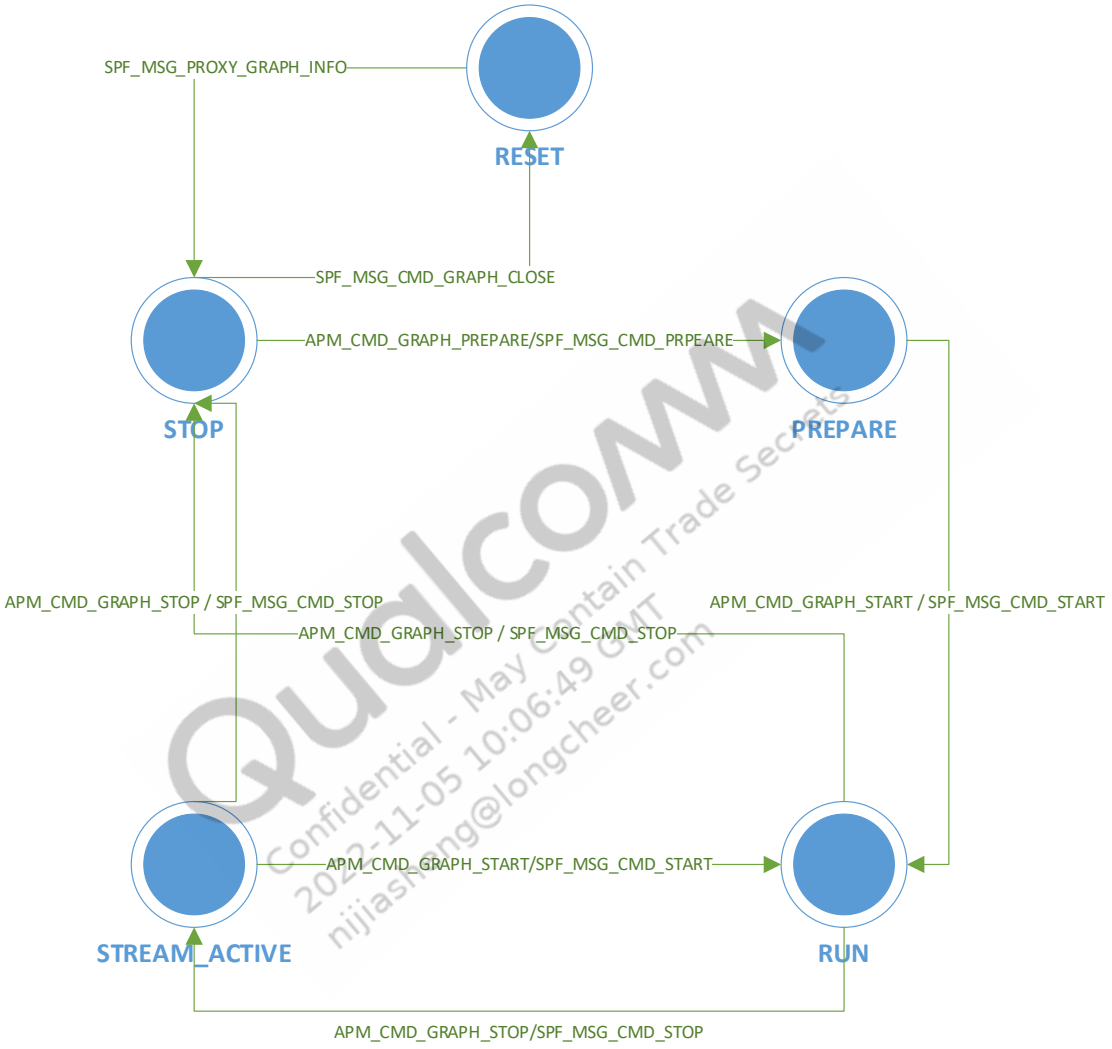
4.2.2 VCPM state machine

The following diagram shows the VCPM state machine.



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4.2.3 VCPM interface and commands

4.2.3.1 Command for VS (modem)

- Timing parameters
  - VCPM_PARAM_ID_TIMING_PARAMS
  - VFR mode, VSID, Tx delivery, Rx start offset, 20/40 ms VFR duration
- Timing reference (For VoLTE)
  - VCPM_PARAM_ID_TIMING_REFERENCES
  - Tx/Rx reference timestamp for voice encoder packet delivery and decoder frame processing start

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- Mailbox configuration
  - VCPM_PARAM_ID_MAILBOX_MEMORY_CONFIG
  - Mailbox circular buffer memory address and size information

4.2.3.2 Command for APPS

- Subgraph configuration
  - VCPM_PARAM_ID_VSID
    - Used to configure VSID property for subgraphs tagged with a voice call scenario ID
  - VCPM_PARAM_ID_VOICE_CONFIG
    - Generic parameter ID that allows the HLOS to set specific configurations/properties to the VCPM
    - Currently, this is used to set the TAG property info. This contains the mapping between tagged modules and their respective instance IDs.
    - Tagging is needed so VCPM and VS can identify the instance ID of modules that they need to interact with directly.
    - VCPM receives this information as part of the graph open command. It caches this as part of its database and also forwards it to VS through the GRAPH_OPEN event.

Module instance ID	Tag ID
miid1	VOICE_MOD_TAG_ID_ENCODER
miid2	VOICE_MOD_TAG_ID_TX_MAILBOX
miid3	VOICE_MOD_TAG_ID_TX_SMART_SYNC
miid4	VOICE_MOD_TAG_ID_DECODER
miid5	VOICE_MOD_TAG_ID_RX_MAILBOX

  - VCPM_PROPERTY_ID_TAG_INFO
    - Used to tag the following modules for configuration from VCPM and VS
      - Voice encoder/decoder
      - Mailbox Tx/Rx
      - Smart Sync
      - G/W/1x/LTE TTY Modules
- Clock scaling
  - VCPM_PARAM_ID_CLK_CTRL
    - Applies a scaling factor on the aggregated module KPPS and bandwidth voting to achieve the desired processor clock frequency and roundtrip delay (RTD) values

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- Module Calibration
  - VCPM_PARAM_ID_CAL_KEYS
    - Used to configure calibration key IDs and corresponding values
    - Pre-defined cal key IDs:
      - Vocoder Class: AMR, EVRC, EVS
      - Sampling rate: 8, 16, 32, 48 kHz
      - Network type: G/W/1x/T/LTE
      - Volume level: Level 1-6
      - Volume boost: Enable/disable
      - Bandwidth extension: Enable/disable
  - VCPM_PARAM_ID_CAL_TABLE
    - Used to configure module calibration for a given subgraph
  - VCPM_PARAM_ID_ACTIVE_CAL_KEYS
    - Used by the calibration tool to query the active calibration key vectors and corresponding values during live voice calls.
- Miscellaneous
  - VCPM_PARAM_ID_NUM_VOL_STEPS
    - Configures the number of volume steps
  - VCPM_PARAM_ID_TTY_MODE
    - Configures TTY mode HCO, VCO, Full
- Voice system ID (VSID)
  - Used to configure voice call graphs for subscription 1 and 2 in the DSDS/DSDA voice call scenario.
  - The following VSID's are available for validating vocoder loopback without a modem with a preconfigured vocoder type and sampling rate configuration:
    - APM_SUB_GRAPH_VSID_SUB1 0x11C05000
    - APM_SUB_GRAPH_VSID_SUB2 0x11DC5000
    - APM_SUB_GRAPH_VSID_VOC_LB_SUB1 0x12006000 (apps only config)
    - APM_SUB_GRAPH_VSID_VOC_LB_SUB2 0x121C6000 (apps only config)
  - Graph design time configuration
    - VSID is configured as part of the subgraph property APM_SUB_GRAPH_PROP_ID_VSID during graph creation from the calibration tool.
    - The graphs are dedicated for a particular subscription and cannot be reused across subs.
  - Graph open time configuration
    - During graph creation, VSID is configured to a default invalid value

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- The actual VSID is configured by the APPS driver after opening the voice call subgraphs.
- The same voice stream/device subgraphs can be used for any of the subscriptions listed above.

4.2.3.3 Command for ADSP

- APM
  - APM receives all graph management and module configuration-related commands from the APPS client
  - However, except for graph open, for all other graph management commands, APM seeks permission from VCPM for the subgraph state transition
- Container
  - Voice call configuration and timing parameters
  - Registers for KPPS, bandwidth and frame done events
  - Manages power manager resource voting for voice call subgraphs
- Modules (via containers)
  - Framework module configuration (for example, smart sync)
  - CKV-based module calibration (for example, for a particular vocoder type or sampling rate combination.)

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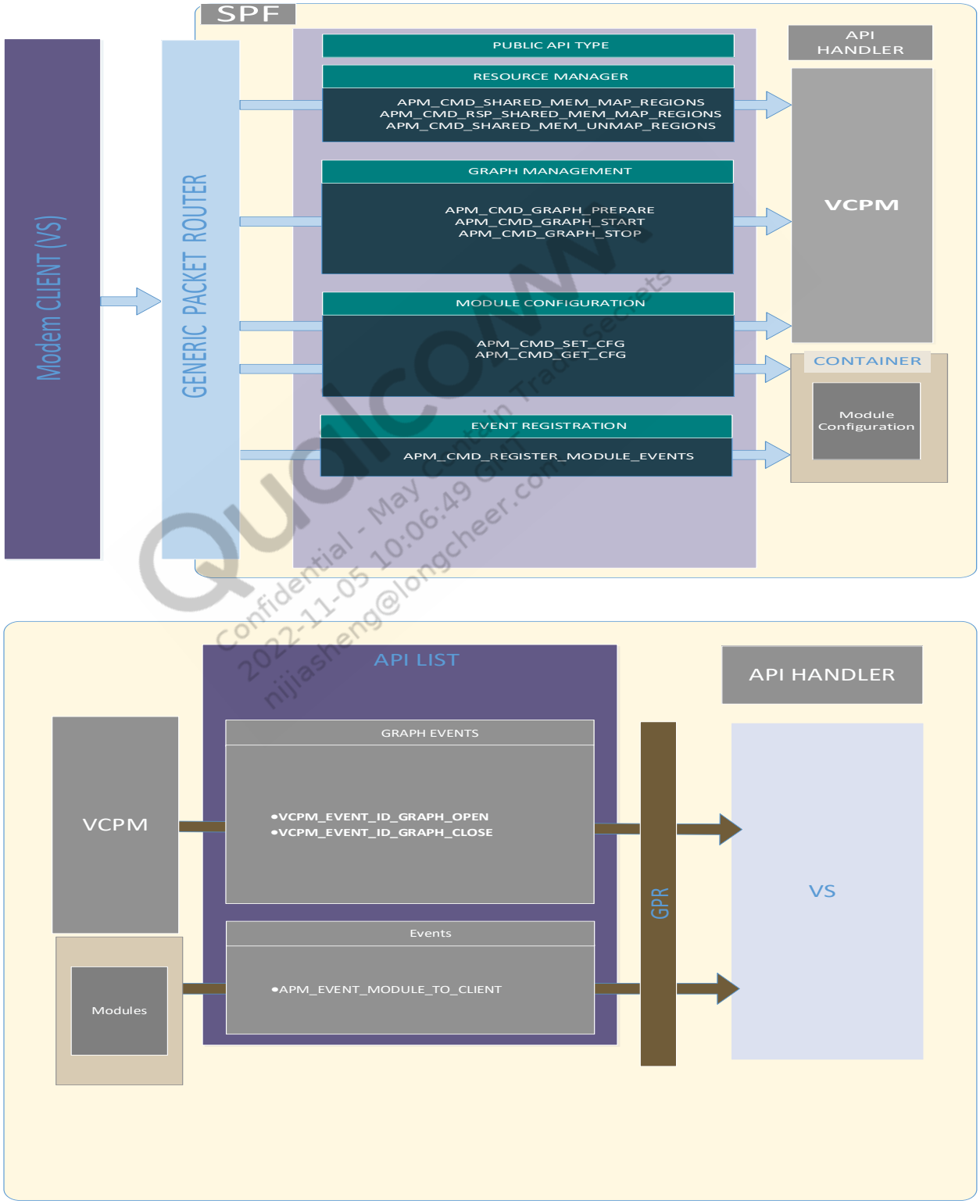
4.2.4 Interface diagrams

4.2.4.1 Interface for VS



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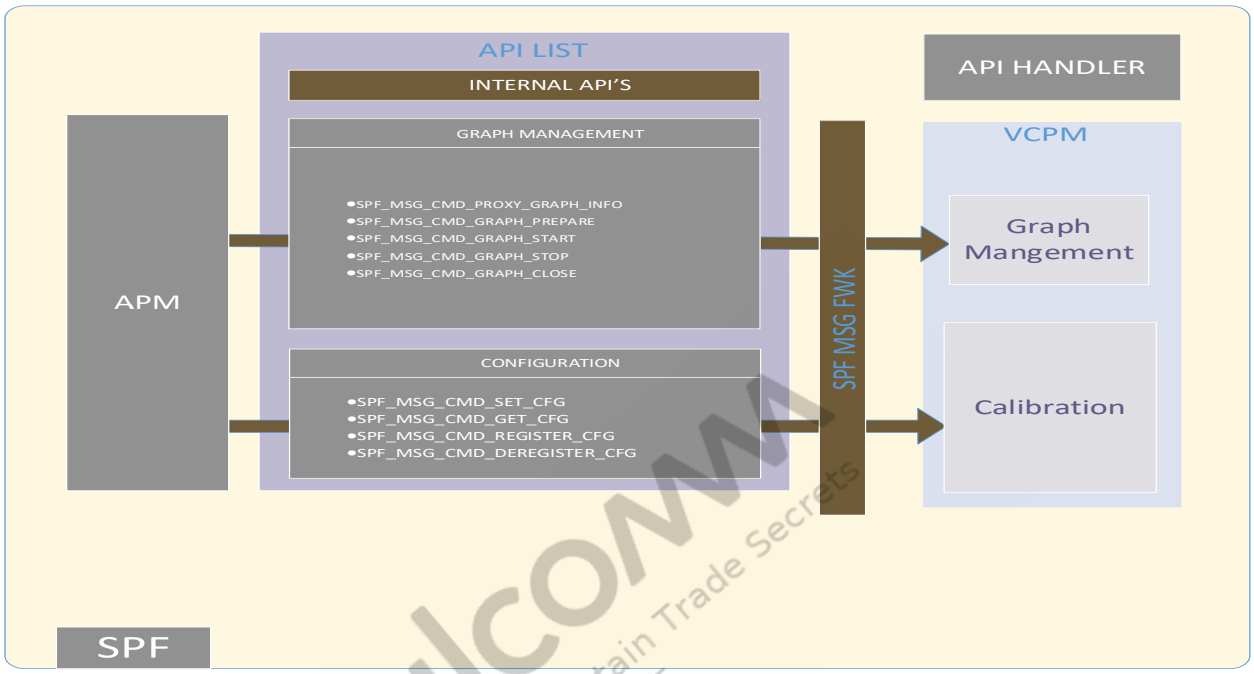
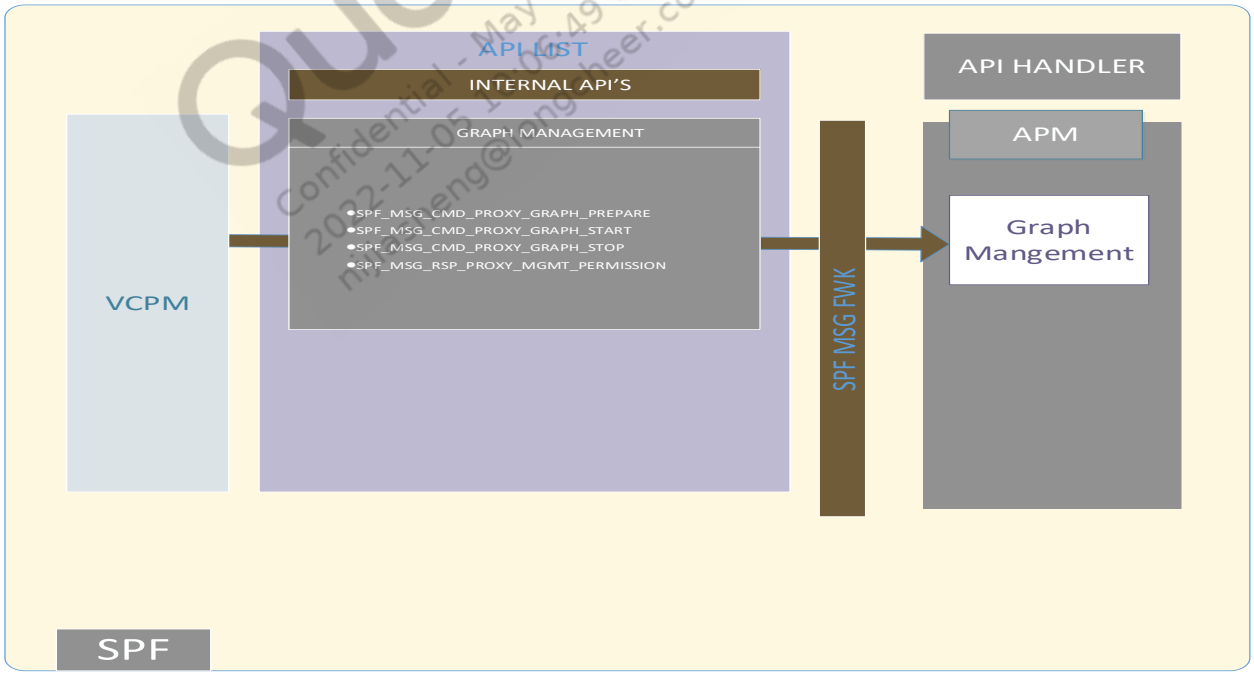
4.2.4.2 Interface for APPS





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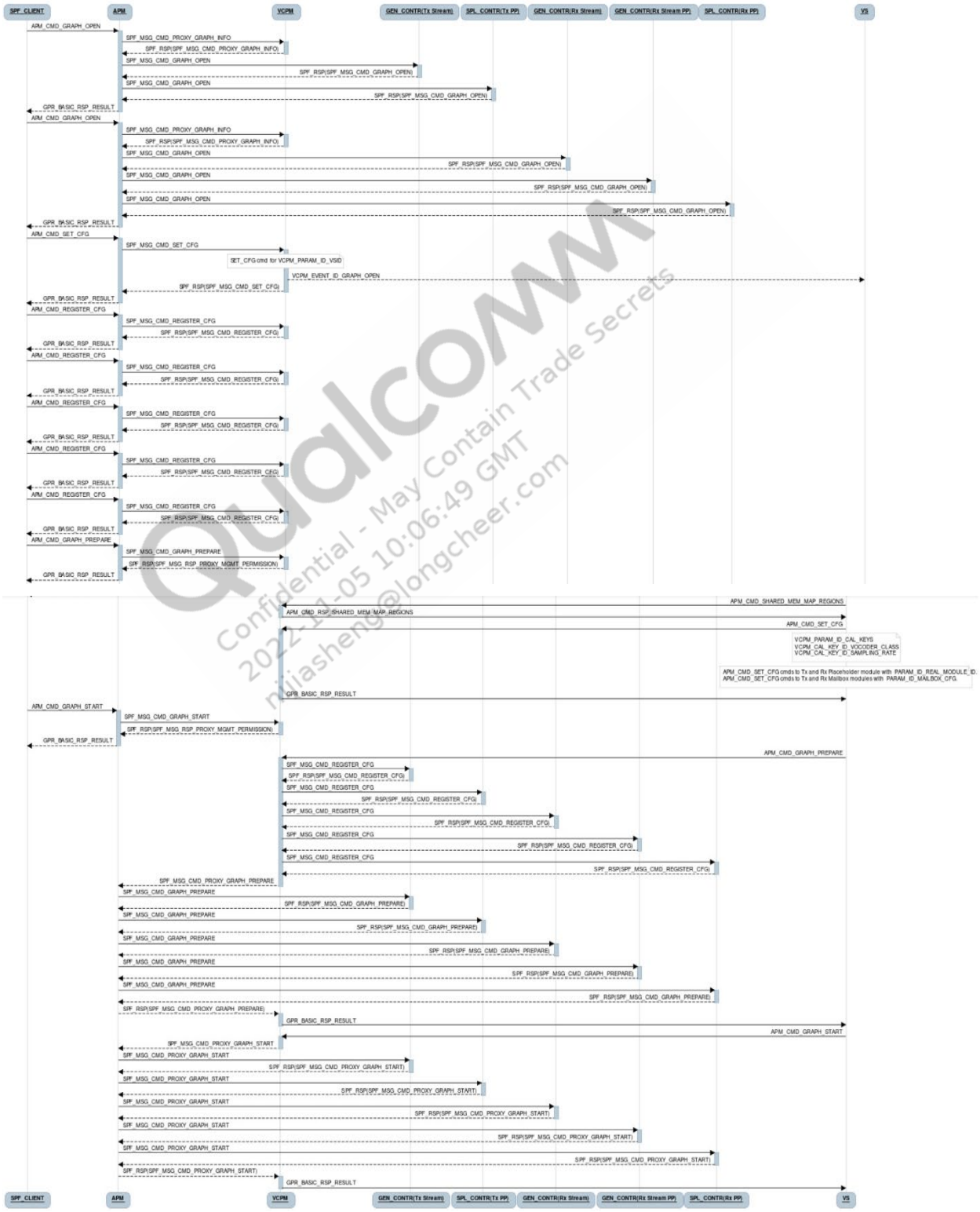
4.3 Call flows

4.3.1 Voice call setup



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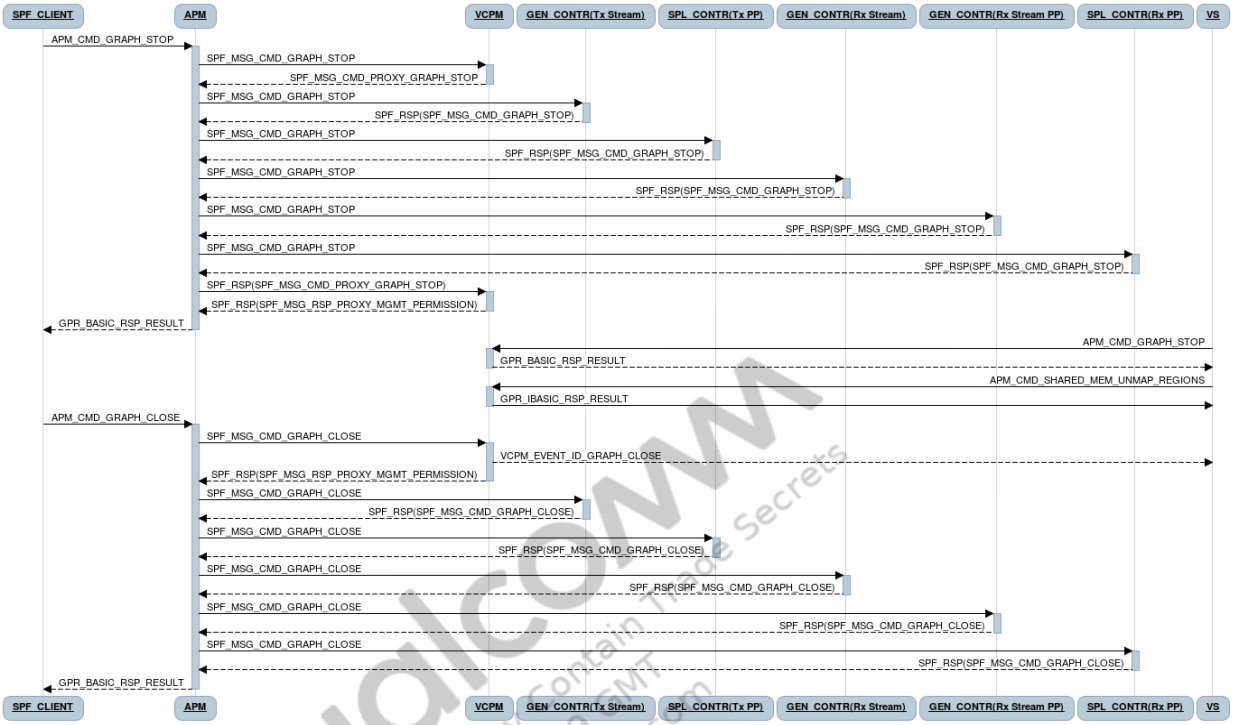
4.3.2 Voice call teardown



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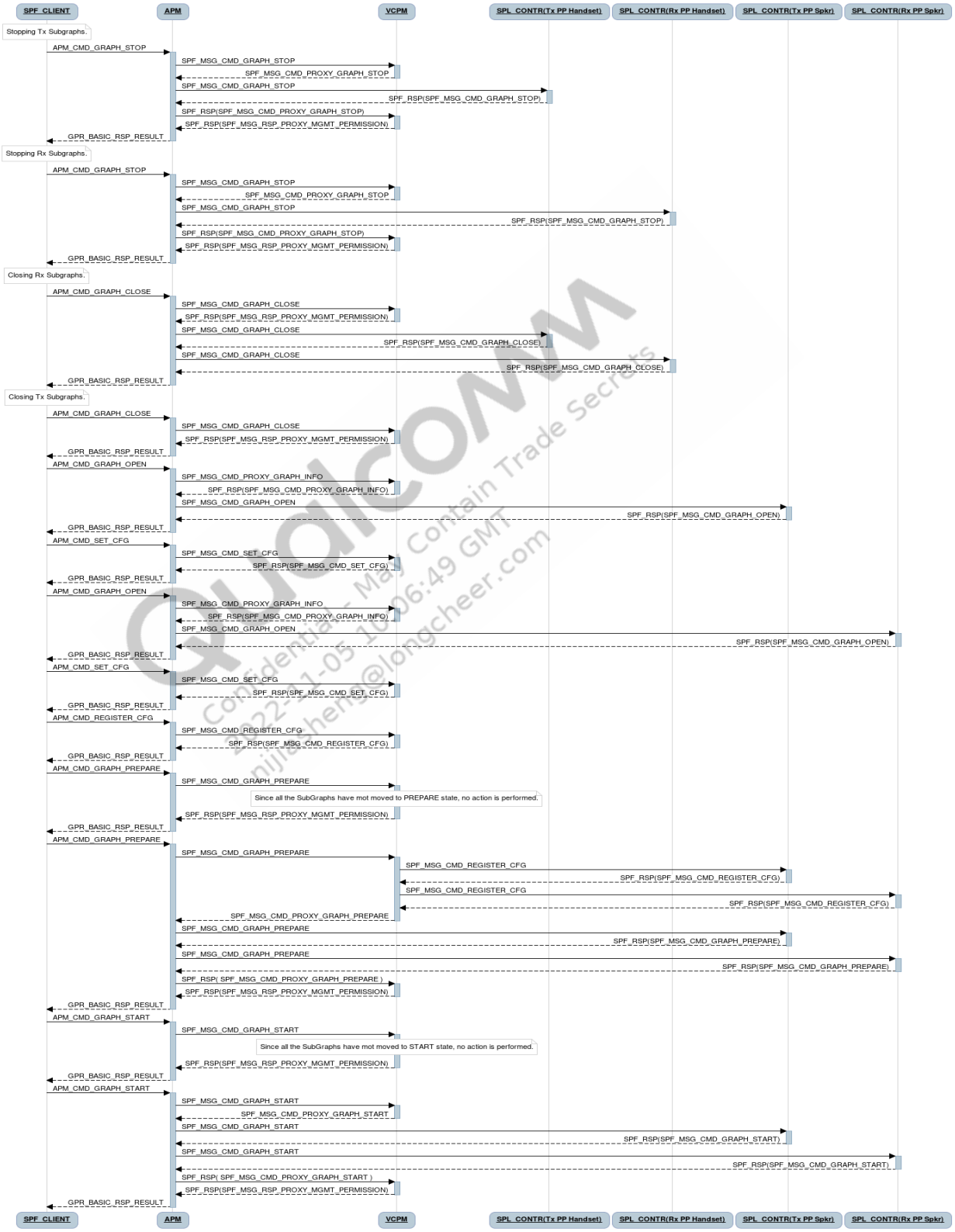
4.3.3 Voice call device switch



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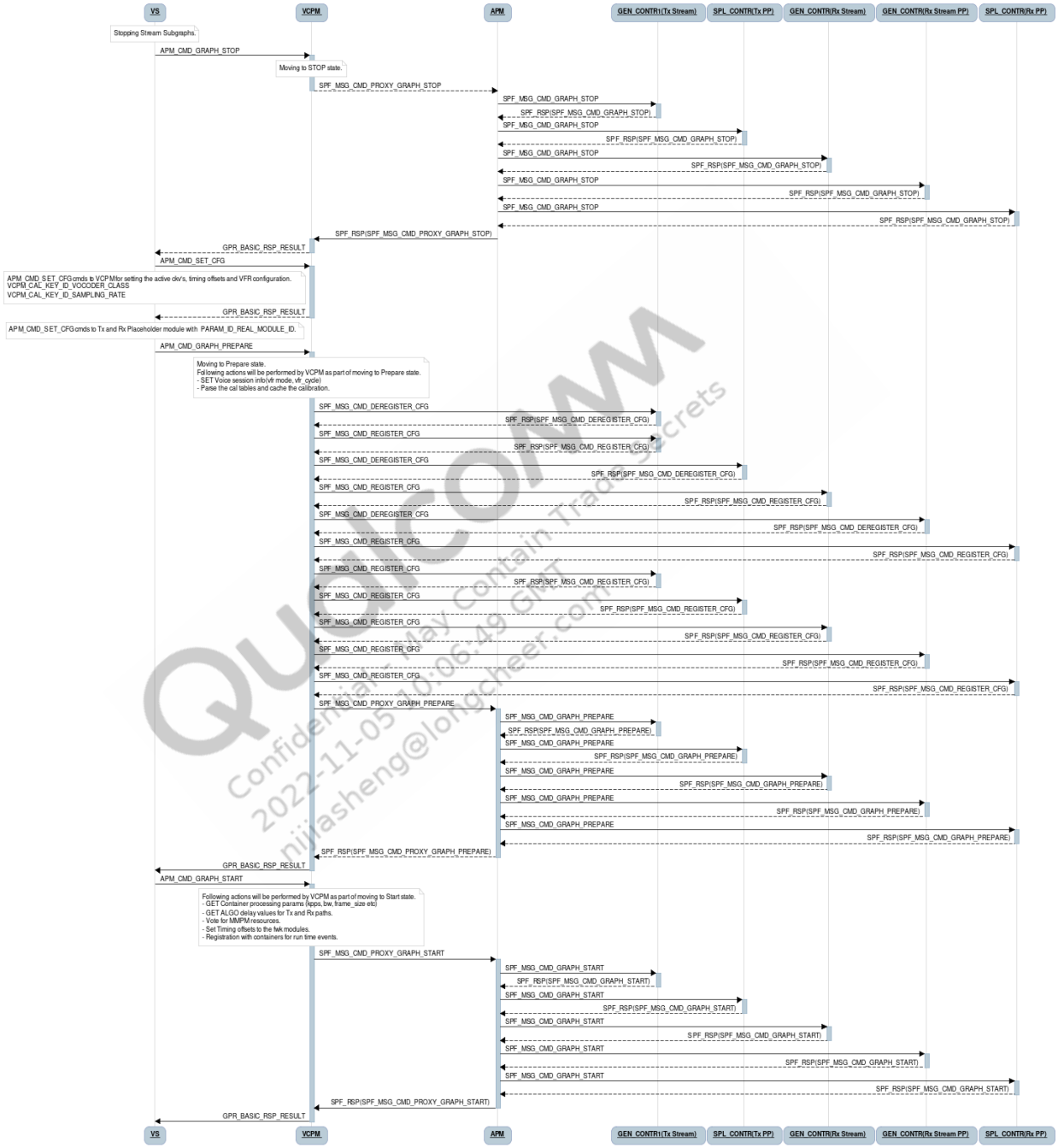
4.3.4 Voice call handover



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5 Logging

5.1 Voice call logging

The following content logged is for a voice call:

- Stream data
  - 0x1586 – Tx input and Rx output
  - 0x158A – Device side logging
  - 0x158B – Stream side logging
  - 0x1804 – Vocoder packets sent from the ADSP (logged at Mailbox Tx)
  - 0x1805 – Vocoder packets received by the ADSP (logged at Mailbox Rx)
  - 0x1914 – Data sent to the modem protocol layers
  - 0x1915 – Data received from the modem protocol layers
- Calibration data
  - 0x188A – Calibration table, Cal keys, Cal params

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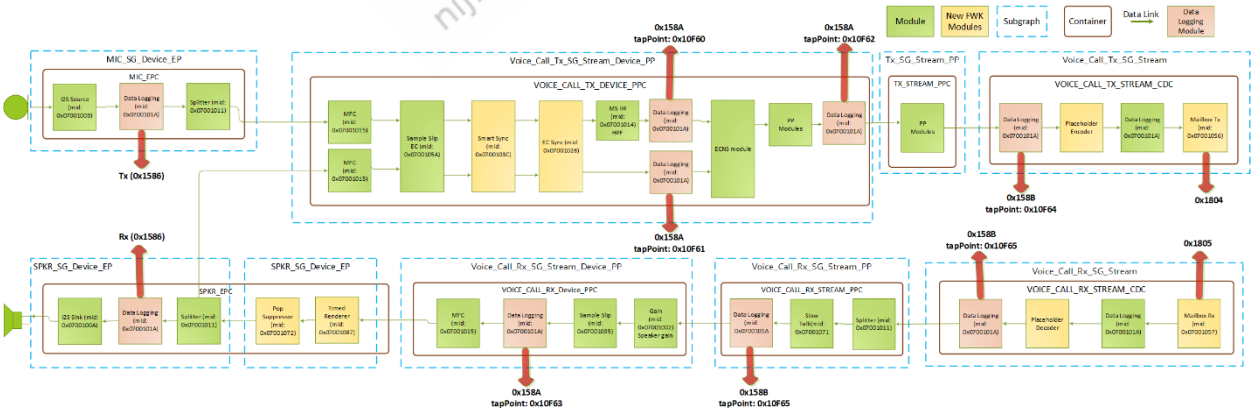
5.1.1 SPF voice call logging diagram



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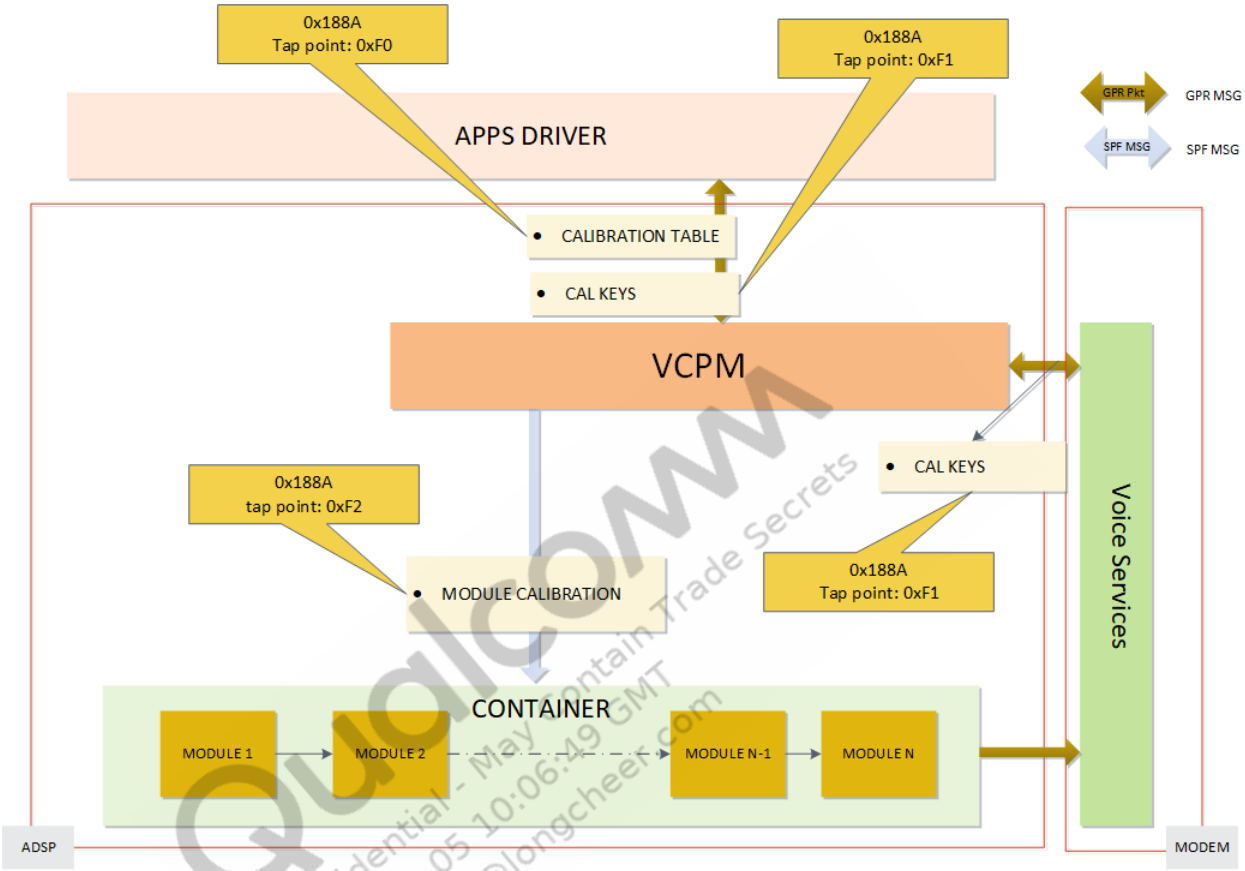
5.1.2 Calibration logging



In this diagram:

- Calibration table
  - The calibration received by the VCPM is validated and added to the cal table list of the subgraph
  - The stored cal table is then logged by the VCPM (tap point 0xF0)
- Cal keys
  - Cal keys are registered with the VCPM for a use case
  - These are logged at the VCPM once set (tap point 0xF1)
- Cal params
  - VCPM searches the calibration corresponding to the active cal keys from the cal table registered and sends the module calibration
  - These params that are sent to the modules are logged in the VCPM (tap point 0xF2)

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6 Customization

6.1 Adding new VSID use case

DSDA and packet loopback are supported on AudioReach. VSID is used for selecting each specific use case, and it is possible to have separated calibration data in ACDB.

This following is an example to make a loopback use case with new VSID:

1. Run a packet loopback.

   APM_SUB_GRAPH_VSID_VOC_LB_SUB1(0x12006000) and APM_SUB_GRAPH_VSID_VOC_LB_SUB2(0x121C6000) are the VSID for loopback. If a stream has been opened with these VSID, it will run a packet loopback inside VCPM without any packet interaction to the modem client.

2. Select VSID GKV for loopback in resourcemanager.xml.

   VSID is set as VSID_default in resourcemanager.xml for all use cases, so it should be modified to set VSID_VOICE1_LB for the loopback use case:

```
// Resourcemanager.xml
    <config_voice>
        <vsid>0xB3000000</vsid>
        <loopbackDelay>1000</loopbackDelay>
        <mode_map>
            <modepair key="0x11C05000" value="0xB3000001"/>
            <modepair key="0x11DC5000" value="0xB3000001"/>
-           <modepair key="0x12006000" value="0xB3000001"/>
+           <modepair key="0x12006000" value="0xB3000004"/>
            <modepair key="0x121C6000" value="0xB3000001"/>
        </mode_map>
    </config_voice>

// Kvh2xml.h
    @h2xmlk_key {VSID}
    @h2xmlk_description {VSID}
 */
 enum Key_VSID {
     VSID_DEFAULT     = 0xB3000001, /*VSID default*/
     VSID_VOICE1      = 0xB3000002, /*VSID 0x11C05000*/
     VSID_VOICE2      = 0xB3000003, /*VSID 0x11DC5000*/
     VSID_VOICE1_LB   = 0xB3000004, /*VSID 0x12006000*/
     VSID_VOICE2_LB   = 0xB3000005, /*VSID 0x121C6000*/
 };
```

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3. Add new graph in ACDB.

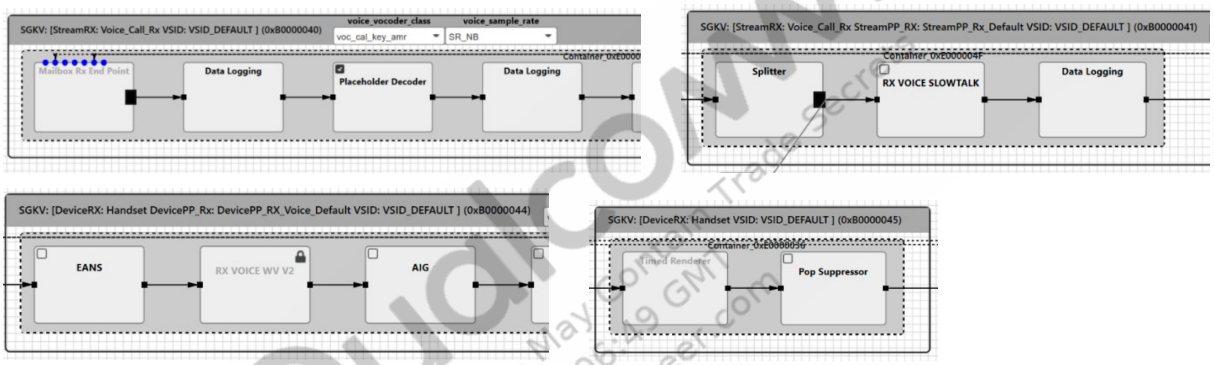
   The following are basic handset call use cases for Rx and Tx. They will be selected if VSID_Default is set from resourcemanager.

   ☑ [StreamRX: Voice_Call_Rx] [DeviceRX: Handset] [DevicePP_Rx: DevicePP_RX_Voice_Default] [StreamPP_RX: StreamPP_Rx_Default] [VSID: VSID_DEFAULT]

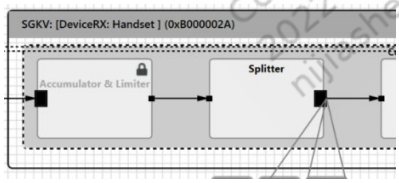
   ☑ [DeviceTX: Handset_Mic] [DevicePP_Tx: Voice_Fluence_SMECNS] [StreamTX: Voice_Call_Tx] [VSID: VSID_DEFAULT]

   VSID_VOICE1_LB is now set in resourcemanager, but there is not a matched graph with that VSID in ACDB. So, it must be added in ACDB.

   The following is the Rx graph for voice call. It has total 5 subgraphs. These 4 subgraphs have VSID GKV, so it should be cloned for a new use case and it should be set VSID_VOICE1_LB as a GKV.



   The following subgraph does not have VSID GKV, so it is not required to clone it. This subgraph will be reused as common.



   Refer to *QACT v8 User Guide* (80-VM407-18) chapter 4.2.3 for instructions on how to create new graph.

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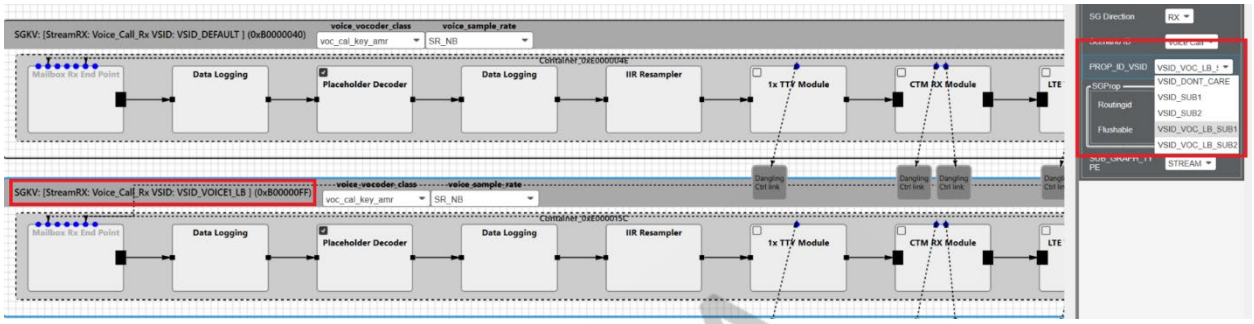
4. Set PROP_ID_VSID in the properties window.

   The following image shows an added new graph with VSID_VOICE1_LB. PROP_ID_VSID in properties window should be set to VSID_VOC_LB_SUB1.



5. Repeat this process for Tx and DSDA. The same approach can be used for the Tx graph as well as VSID_VOICE1 and VSID_VOICE2 for the DSDA use case.

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A References

A.1 Related documents

Title	Number
Qualcomm Technologies, Inc.	
AudioReach Signal Processing Framework Overview	80-VN500-3
CASA LA Architecture	80-VN500-7
QACT v8 User Guide	80-VM407-18

A.2 Acronyms and terms

Acronym or term	Definition
ACDB	Audio Calibration Database
CASA	Common Audio Software Architecture
PAL	Qualcomm Audio Layer
QACT	Qualcomm Audio Calibration Tool

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